

Laboratory Activity

User Interface Design Principles

Course	Topic	Duration	Mode
Software Architecture	User Interface (UI) Design	2 hours	Pair Work

Learning Objective

By the end of this activity, students will be able to acknowledge and apply the following UI design principles when creating a user interface design:

- Consistency — uniform patterns, terminology, and behavior across the interface
- Usability — ease of learning and efficiency of use
- Navigation — clear, predictable pathways through the system
- Visual Appeal — aesthetically pleasing and purposeful visual design
- Interoperability — ability to work across devices, platforms, and systems
- Performance — responsiveness and efficiency of the interface
- Accessibility — usability for people with diverse abilities and needs

Materials / Tools Needed

- Laptop/PC with internet access
- Design tool (choose one): Figma, Adobe XD, or Canva
- Access to at least 2 existing mobile or web applications (for evaluation task)
- Rubric sheet (provided at the end of this document)
- Sticky notes or a digital whiteboard (Miro/FigJam), optional but recommended for Task 2

Background

A well-architected software system is not judged only by its backend logic — the User Interface (UI) is the layer through which users experience and evaluate the entire system. Poor UI design can make even a technically excellent system feel broken, while thoughtful UI design can make a simple system feel powerful and trustworthy. This activity walks students through evaluating an existing interface and then designing their own, applying seven core UI design principles at every step.

Task	Activity	Duration
1	Orientation & Grouping	10 mins
2	Heuristic Evaluation of an Existing App	30 mins
3	Requirements Definition for New UI	15 mins
4	Low-Fidelity Wireframing	30 mins
5	High-Fidelity Prototype (Partial)	25 mins

Task	Activity	Duration
6	Peer Review & Principle Checklist	15 mins
7	Presentation & Wrap-up	15 mins
	Total	~2 hours

Task 1 — Orientation & Grouping (10 minutes)

1. Form pairs (or work individually if the class size requires it).
2. Each pair selects one problem domain for their UI design project. Choose one:
 - A campus event registration app
 - A local food delivery app
 - A barangay/community service request portal
 - A student attendance and grading portal
3. Write down your chosen domain and a one-sentence purpose statement.

Task 2 — Heuristic Evaluation of an Existing App (30 minutes)

Before designing, learn to see UI principles in action by critiquing something that already exists.

1. Choose one existing app or website related (even loosely) to your domain (e.g., if you chose food delivery, evaluate GrabFood, Foodpanda, or a similar app).
2. Explore the app for 10 minutes, performing 3 common tasks a user would do (e.g., search, add to cart, check out).
3. Fill out the UI Evaluation Table below, rating each principle from 1 (poor) to 5 (excellent) and giving a short justification.

Principle	Rating (1–5)	Observation / Justification
Consistency		
Usability		
Navigation		
Visual Appeal		
Interoperability		
Performance		
Accessibility		

Guiding questions:

- Do buttons, icons, and colors mean the same thing everywhere in the app? (Consistency)
- Could a first-time user complete a task without instructions? (Usability)

- Can users tell where they are and how to get back? (Navigation)
- Does the interface feel visually organized and pleasant, not cluttered? (Visual Appeal)
- Does it work well across screen sizes/devices, or integrate with other services (e.g., maps, payment gateways)? (Interoperability)
- Does it respond quickly, with minimal lag or loading delay? (Performance)
- Does it support screen readers, sufficient color contrast, text resizing, or alt text? (Accessibility)

Task 3 — Requirements Definition for New UI (15 minutes)

Using insights from Task 2, define the requirements for your own UI design project.

1. List 3–5 core user tasks your interface must support (e.g., "register for an event," "view event schedule," "receive confirmation").
2. Identify your target users, including at least one accessibility consideration (e.g., users with low vision, users on slow internet connections, older adult users).
3. Write one guiding principle statement per UI principle, e.g.:

Consistency: All primary action buttons will use the same color, shape, and placement across every screen.

Principle	Design Commitment for This Project
Consistency	
Usability	
Navigation	
Visual Appeal	
Interoperability	
Performance	
Accessibility	

Task 4 — Low-Fidelity Wireframing (30 minutes)

1. Sketch (on paper or in your chosen tool) at least 3 core screens based on your Task 3 tasks.
Examples:
 - Home / landing screen
 - A form or input screen (e.g., registration, order form)
 - A confirmation or results screen
2. On each wireframe, label where each UI principle is being applied. Example annotations:
 - "Consistent header across all screens" → Consistency
 - "Breadcrumb trail" → Navigation
 - "Large tap targets, min 44x44px" → Accessibility
3. Include a simple navigation flow diagram showing how a user moves between your 3 screens (boxes and arrows are sufficient).

Task 5 — High-Fidelity Prototype, Partial (25 minutes)

1. Choose one screen from Task 4 and turn it into a higher-fidelity mockup using your design tool of choice (color, typography, real UI components).
2. Apply at least:
 - 1 accessibility feature (e.g., color contrast $\geq 4.5:1$, alt text placeholders, legible font size $\geq 14\text{px}$)
 - 1 performance consideration (e.g., note where lazy loading, image compression, or skeleton loaders would apply)
 - 1 interoperability consideration (e.g., responsive layout notes for mobile vs. desktop)
3. Export or screenshot your mockup for submission.

Task 6 — Peer Review & Principle Checklist (15 minutes)

1. Exchange your wireframes/prototype with another pair.
2. Using the Peer Review Checklist below, evaluate your partner's design and give at least one improvement suggestion per principle.

Principle	Present? (✓ / ✗)	Suggestion for Improvement
Consistency		
Usability		
Navigation		
Visual Appeal		
Interoperability		
Performance		
Accessibility		

3. Return the checklist to the original pair for their reflection.

Task 7 — Presentation & Wrap-up (15 minutes)

Each pair presents (2–3 minutes) covering:

1. Their chosen domain and target users
2. How each of the 7 principles was applied (briefly, one sentence each)
3. One piece of peer feedback they plan to act on

Deliverables (to submit)

1. Completed UI Evaluation Table (Task 2)

2. Completed Requirements Table (Task 3)
3. Wireframe sketches with principle annotations (Task 4)
4. Navigation flow diagram (Task 4)
5. One high-fidelity screen mockup (Task 5)
6. Completed Peer Review Checklist (Task 6)

Grading Rubric (100 points)

Criteria	Excellent	Satisfactory	Needs Improvement
Consistency (10 pts)	Uniform components, colors, and terminology throughout	Mostly consistent, minor lapses	Inconsistent patterns across screens
Usability (15 pts)	Tasks are intuitive and efficient to complete	Usable but requires some guidance	Confusing or inefficient task flow
Navigation (15 pts)	Clear, predictable, and well-mapped navigation flow	Navigation mostly clear, minor gaps	Navigation unclear or incomplete
Visual Appeal (15 pts)	Polished, purposeful, visually balanced design	Acceptable visual design	Cluttered or unpolished
Interoperability (10 pts)	Clear responsive/cross-platform considerations	Some consideration shown	Not addressed
Performance (10 pts)	Thoughtful performance notes/optimizations identified	Basic performance awareness	Not addressed
Accessibility (15 pts)	Concrete accessibility features applied and justified	Some accessibility features present	Accessibility not considered
Documentation & Presentation (10 pts)	Clear, complete, well-presented deliverables	Mostly complete	Incomplete or unclear

Total: 100 points

Reflection Questions

(for submission)

1. Which of the 7 principles was hardest to apply in your design, and why?
2. How might ignoring accessibility affect real-world users of your app?
3. How does performance as a UI concern relate to backend/software architecture decisions?
4. If you had one more hour, which principle would you improve further, and how?

End of Laboratory Activity